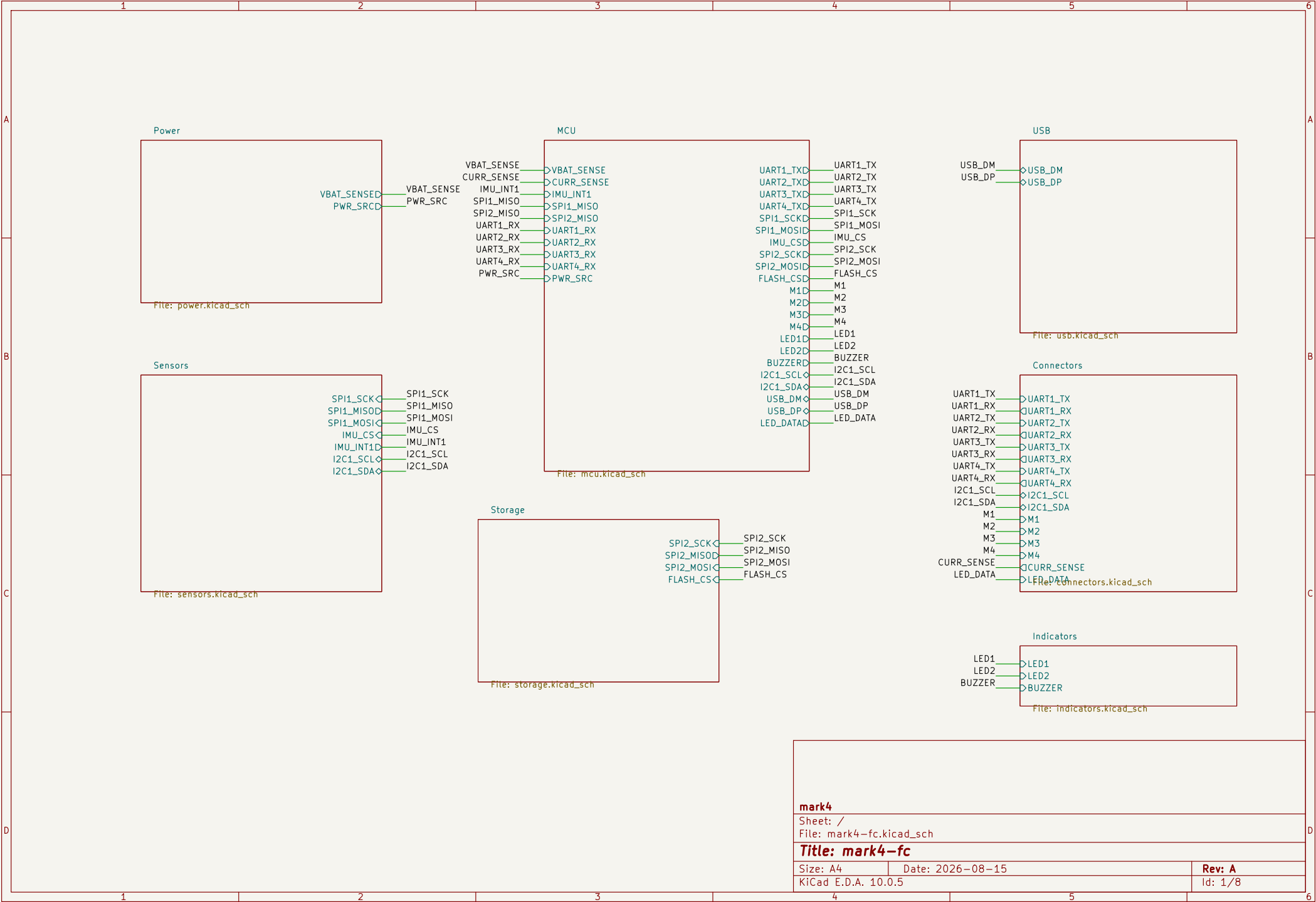




mark4

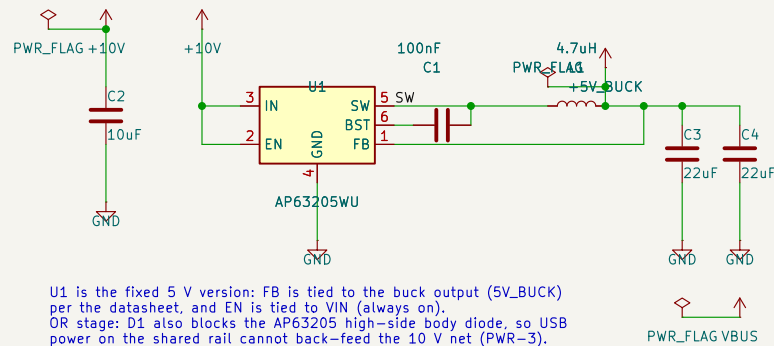
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File: mark4-fc.kicad_sch

Title: mark4-fc

Size: A4 Date: 2026-08-15
KiCad E.D.A. 10.0.5

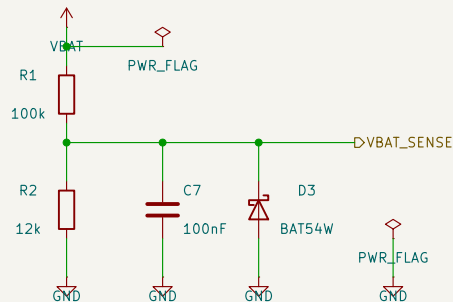
Rev: A
Id: 1/8

10 V BEC input and 5 V buck (PWR-1)



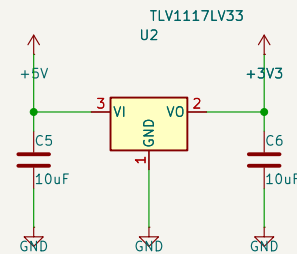
U1 is the fixed 5 V version: FB is tied to the buck output (5V_BUCK) per the datasheet, and EN is tied to VIN (always on).
OR stage: D1 also blocks the AP63205 high-side body diode, so USB power on the shared rail cannot back-feed the 10 V net (PWR-3).
The board is fed from the ESC BEC and never from raw VBAT: motor braking spikes on a 6S pack can exceed the 32 V input rating of the AP63205, and the BEC absorbs them. Measured source: T-Motor Velox 45A V2, 10 V BEC.

VBAT sense to ADC, raw battery at the ESC pad group (PWR-4)

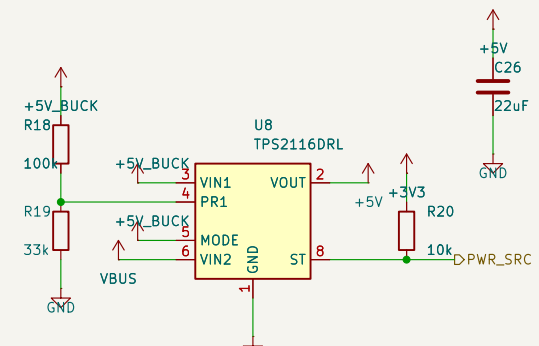


The divider now hangs on the raw battery instead of a regulated rail, so pack voltage is directly visible; the old input-rail divider could not see it.
 $R1/R2 = 100k/12k$, ratio $12/112 = 0.10714$.
Tap = 2.70 V at 25.2 V (6S max), 1.80 V at 16.8 V (4S max), and 3.3 V (VDDA) is reached only at 30.8 V, so any plausible pack transient stays inside the ADC range.
Drain 0.22 mA continuous while the pack is plugged; source impedance 10.7 k, fine because C7 sits at the pin and the conversion runs at about 1 Hz with a long sampling time.
D3 clamps negative excursions: DS11853 Table 60 allows zero negative current injection on PC0..PC3.

3V3 LDO (PWR-2)



5 V active OR-ing / power mux (PWR-3)



U8 replaces the two Schottky ORing diodes: 40 mOhm typical (60 mOhm over temperature) instead of 300 mV of drop, and it blocks in BOTH legs, so USB 5 V cannot back-drive the AP63205 output through its high-side body diode. MODE tied to VIN1 selects priority mode: the BEC wins whenever it is there, so plugging or unplugging USB never disturbs a board that is already flying. R18/R19 set the switchover threshold at $VREF * 133/33 = 4.03$ V on VIN1 ($VREF = 0.92$ to 1.08 V, so 3.7 to 4.35 V worst case; the buck holds 5 V $\pm 2\%$, so no parasitic switchover).
With VIN1 dead (bench, USB only) MODE is low and PR1 is low, and the truth table passes the higher of the two inputs: the board runs on USB with nothing to command.
Switchover is BREAK-before-make, $t_{SW} = 8$ us at 5 V: C5 + C26 keep the droop near 145 mV at 400 mA.
VIN abs max is 6 V (5.5 V recommended). Unlike the Schottky stage, this node is no longer indifferent to a USB overvoltage. No extra TVS was added for that (2026-08-17): U7 (USBLC6-25C6) already clamps VBUS at VBR 6.1 V, and no passive part can hold 5.25 V and cap at 6 V – only an active OVP switch could, and a compliant UFP source never leaves 5 V. See PWR-3.

mark4

Sheet: /Power/
File: power.kicad_sch

Title: mark4-fc - Power

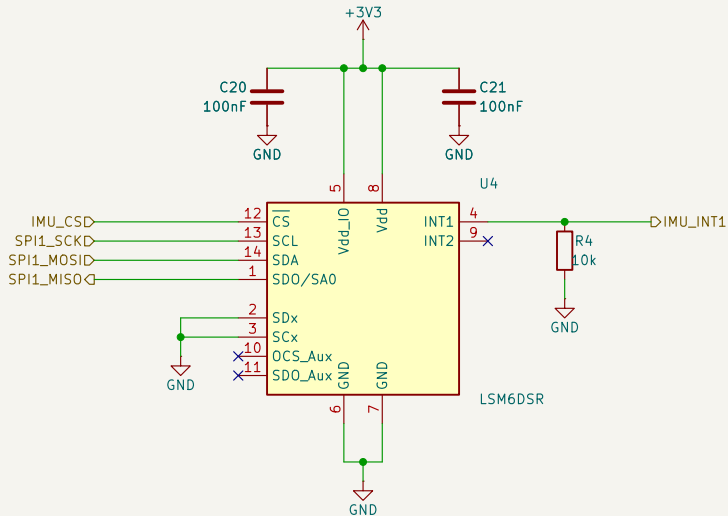
Size: A4 Date: 2026-08-15

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Rev: A

Id: 2/8

IMU – LSM6DSR on SPI1, dedicated CS (SNS–1)

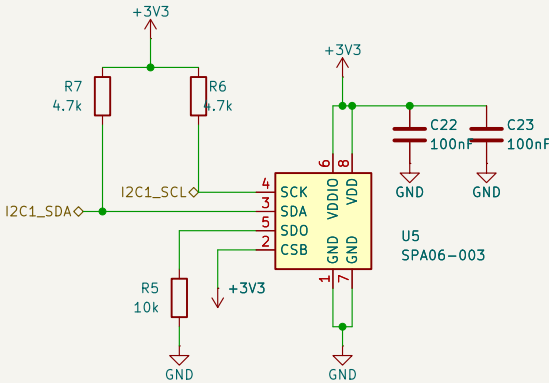


SNS–1 SPI Mode 1: SDx and SCx tied to GND select the 4–wire SPI primary interface. OCS_Aux and SDO_Aux stay UNCONNECTED (soldered, no net) – they carry internal pull–ups, so grounding them burns current and is out of spec. INT2 unused, no–connect flag. SPI1 clock 10 MHz max; F722 SPI1 prescaler /16 gives 6.75 MHz. FIRMWARE, first writes after reset: I2C_disable = 1 (CTRL4_C) and I3C_disable = 1. With CS high the I2C/I3C slave is otherwise live on the SPI lines.

SNS–2 INT1 is an interface strap sampled at IMU power–up (high = I3C–only mode). NO pull–up on this net, EVER. R4 10k pull–down is the optional safe default; the MCU pin stays input / low until the IMU is up. Routing INT1 to an EXTI–capable pin is mandatory, using it is optional (the loop may stay timer–paced).

SNS–5 IMU placement: board center, stiffest area, axes aligned with the board edges and marked on silkscreen, continuous ground plane underneath. C20 / C21 100 nF at the Vdd_IO / Vdd pins. The ESC silicone grommets are the vibration isolation (MEC–1).

Barometer – SPA06–003 on I2C1, addr 0x76 (SNS–3)



SNS–4 baro placement: pressure port facing UP, no silkscreen and no conformal coating on the port, room for an open–cell foam patch over it, away from the regulators (thermal gradient = drift) and away from the prop–wash board edges.

SNS–3 rate ceiling from the datasheet (Rate × Time < 1 s); 25 Hz allows 16x pressure oversampling, 50 Hz caps it at 8x. CSB tied to +3V3 (= VDDIO level) selects I2C. SDO to GND through R5 10k gives address 0x76; a resistor rather than a short keeps SDO–as–interrupt available.

R6 / R7 4.7k are the ONLY I2C1 pull–ups on the board. They live on this sheet because the barometer is the primary I2C user; the GPS connector (CON–2) shares the same bus and must NOT carry a second pair.

mark4

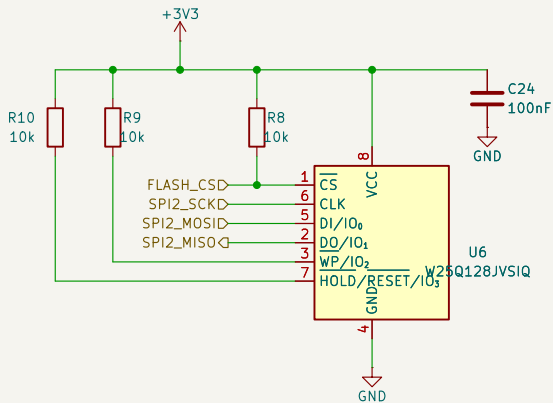
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File: sensors.kicad_sch

Title: mark4–fc – Sensors

Size: A3 Date: 2026–08–15
KiCad E.D.A. 10.0.5

Rev: A
Id: 3/8

Blackbox storage – W25Q128JV on its own SPI2 bus (STO–1)



STO–1 strapping, R8 10k on /CS: the die has no internal pull–up and the datasheet requires /CS to track VCC at power–up, when the MCU pin is still high–Z. R9 / R10 10k on /WP (IO2) and /HOLD (IO3): NEVER hard ties – the IQ order code ships with QE=1, which repurposes both pins as IO2/IO3, and the datasheet explicitly forbids direct supply or ground ties in that state. Own SPI bus, never shared with the IMU: logging writes must not contend with sensor reads. Plain SPI only, QUADSPI dropped (STO–3).

FOOTPRINT: the KiCad symbol default (SOIC–8 3.9x4.9 mm) is WRONG for the JVS1Q 208–mil wide body. Overridden to mark4–fc:SOIC–8_5.28x5.28mm_P1.27mm.

STO–5 throughput – why the bus is never the constraint, and the FIRMWARE rule. Required sustained rate is 32.5 kB/s (65–byte record at 500 Hz, STO–2). The W25Q128JV programs a 256–byte page in 0.7 ms typ / 3 ms max = 366 kB/s typ and 85 kB/s worst case, both above the requirement. SPI clock is irrelevant to writes (50 MHz ceiling for plain 03h reads, 27 MHz chosen on SPI2). What matters is ERASE latency: a 4 KB sector erase stalls the die 45 ms typ, up to 400 ms max = 13 kB of backlog at the required rate. FIRMWARE MUST therefore buffer records in a RAM ring of 16 KB or more (cheap out of 256 KB) and erase AHEAD of the write head.

Sizing (STO–2): 16 MB is about 8 minutes of armed logging: logging only while armed is the intended policy. W25Q256JVS1Q (32 MB, same footprint) MAY be substituted at order time. Retrieval over USB CDC (STO–4), no SD card.

mark4

Sheet: /Storage/
File: storage.kicad_sch

Title: mark4–fc – Storage

Size: A3

Date: 2026–08–15

Rev: A

KiCad E.D.A. 10.0.5

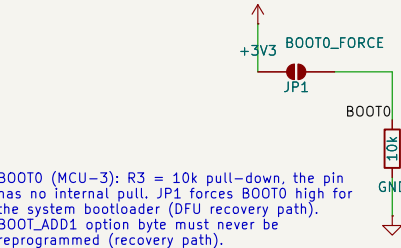
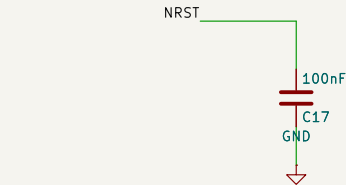
Id: 4/8

MCU – STM32F722RET6 (MCU–1). Pin assignment per docs/aio–board–pinmap.md

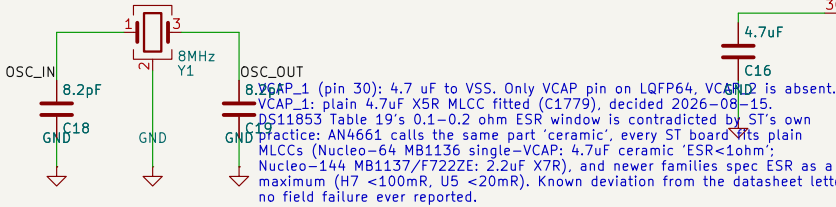
Decoupling per the pin map summary: C8–C11 = one 100 nF per VDD pin (19, 32, 48, 64), each placed at its pin on the PCB. C12 = 4.7 uF bulk on 3V3 near the MCU. C13 = 100 nF on VBAT (pin 1, tied to +3V3; no coin cell, no RTC backup).

VDDA (pin 13) is tied to +3V3 directly – NO series resistor: VDDA is the ADC reference, an IR drop is a reference offset. C14 100 nF + C15 1 uF at the pin (DS11853 Figure 28). VSSA (pin 12) joins the ground plane at one point near the MCU.

NRST (pin 7): 100 nF at the pin (DS11853 Figure 47). Internal pull–up 30–50k, no external one. Not 5 V tolerant (4.0 V max) – exposed on J1 pin 5.

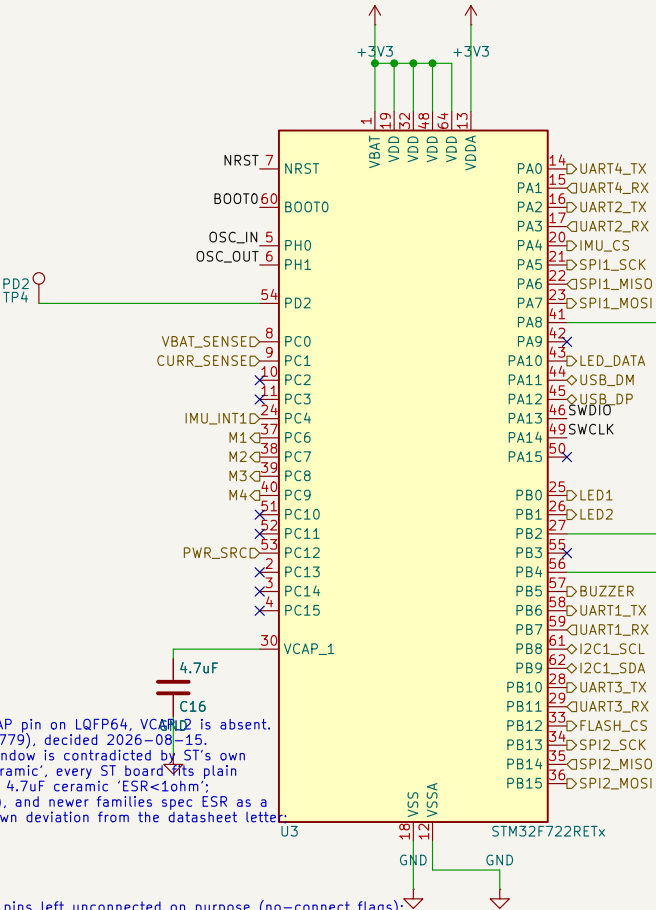
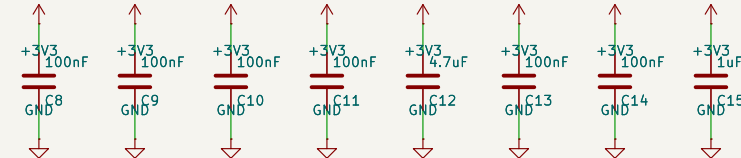


BOOT0 (MCU–3): R3 = 10k pull–down, the pin has no internal pull. JP1 forces BOOT0 high for the system bootloader (DFU recovery path). BOOT_ADD1 option byte must never be reprogrammed (recovery path).

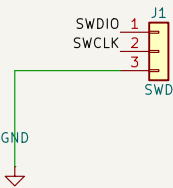


HSE 8 MHz crystal (MCU–4): PLL /4 x216 /2 = 216 MHz sysclk, /9 = exact 48 MHz USB clock. C18/C19 = 2 x (CL 9 pF – 5 pF stray) = 8.2 pF, NPO only; trim 6.8/10 pF on the bench if the layout runs long. Refs chosen 2026–08–15: C1685, 8.2 pF COG 50 V 0603 (exact computed value). Tolerance C = +/-0.25 pF inferred from the FH part code; confirm on the datasheet at order time if the AN2867 margin matters. Layout: guard ground ring around Y1 and its caps stitched to the MCU VSS, PH0/PH1 loop as short as possible, nothing routed under the crystal body.

Free pins left unconnected on purpose (no–connect flags): PC13, PC14, PC15 (backup domain, no LSE fitted). PC2, PC3 (spare ADC inputs). PA9 (OTG_FS_VBUS – sensing stays disabled, firmware forces B–valid). PA10 (OTG_FS_ID, device–only board). PA15, PC10, PC11 (UART escape route). PC12. They stay reachable at the package if a respin ever needs them.



SWD header (MCU–2 / CON–4): 2.54 mm 1x6 through–hole, mark1 Prog pinout 1 = 3V3, 2 = SWCLK, 3 = GND, 4 = SWDIO, 5 = NRST, 6 = SWO (PB3); NOT assembled: hand–populated with 3 pins (GND, SWCLK, SWDIO); all 6 positions are routed so the 1:1 cable option stays open. No BOM line. SWD nets stay local to this sheet.



Test points (pin map section 4 recommendation): TP1 = PA8 (MCO_1, clock bring–up probe), TP2 = PB2, TP3 = PB4 (NJTRST at reset), TP4 = PD2 (only GPIOD pin).

Not on this sheet: the I2C1 4.7k pull–ups on PB8/PB9 (sensor sheet), the 10k pull–up on FLASH_CS/PB12 (storage sheet, the W25Q /CS must track VCC at power–up), the VBAT divider and its Schottky feeding VBAT_SENSE (power sheet, PWR–4), and the USB 90 ohm pair plus ESD array on USB_DM/USB_DP (USB sheet, CON–5, no series resistors). PA4 and PA5 are TTA pins (3.3 V max): nothing 5 V may ever touch IMU_CS or SPI1_SCK.

mark4

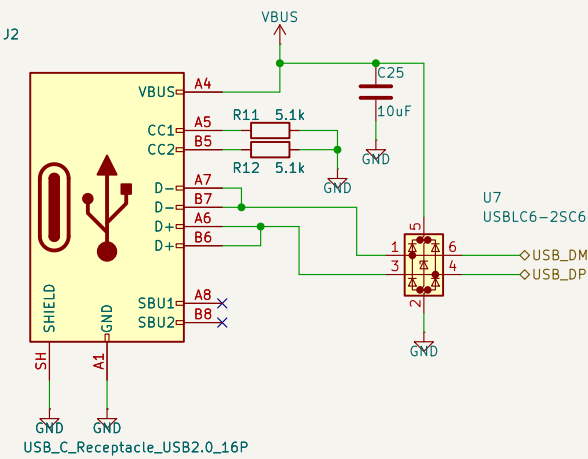
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File: mcu.kicad_sch

Title: mark4–fc – MCU

Size: A3 Date: 2026–08–15
KiCad E.D.A. 10.0.5

Rev: A
Id: 5/8

USB (CON-5) – USB-C 2.0 receptacle, UFP only



- Notes:
- Route USB_DP / USB_DM as a 90 ohm differential pair at layout (CON-5).
 - No series resistors on D+/D- and no VBUS divider; firmware forces B-valid (pinmap PA9 row).
 - R11 / R12: one 5.1k pulldown per CC pin, NEVER shared – this is a UFP-only port.
 - C25 is local VBUS bulk near the receptacle; VBUS is the bench supply path (PWR-3).
 - OR-ed with the buck output on the Power sheet.
 - SBU1 / SBU2 unused (USB 2.0 only); shell and GND pins go straight to the GND plane.
 - U7 sits between the receptacle and the MCU (flow-through): I/O1 = D-, I/O2 = D+.

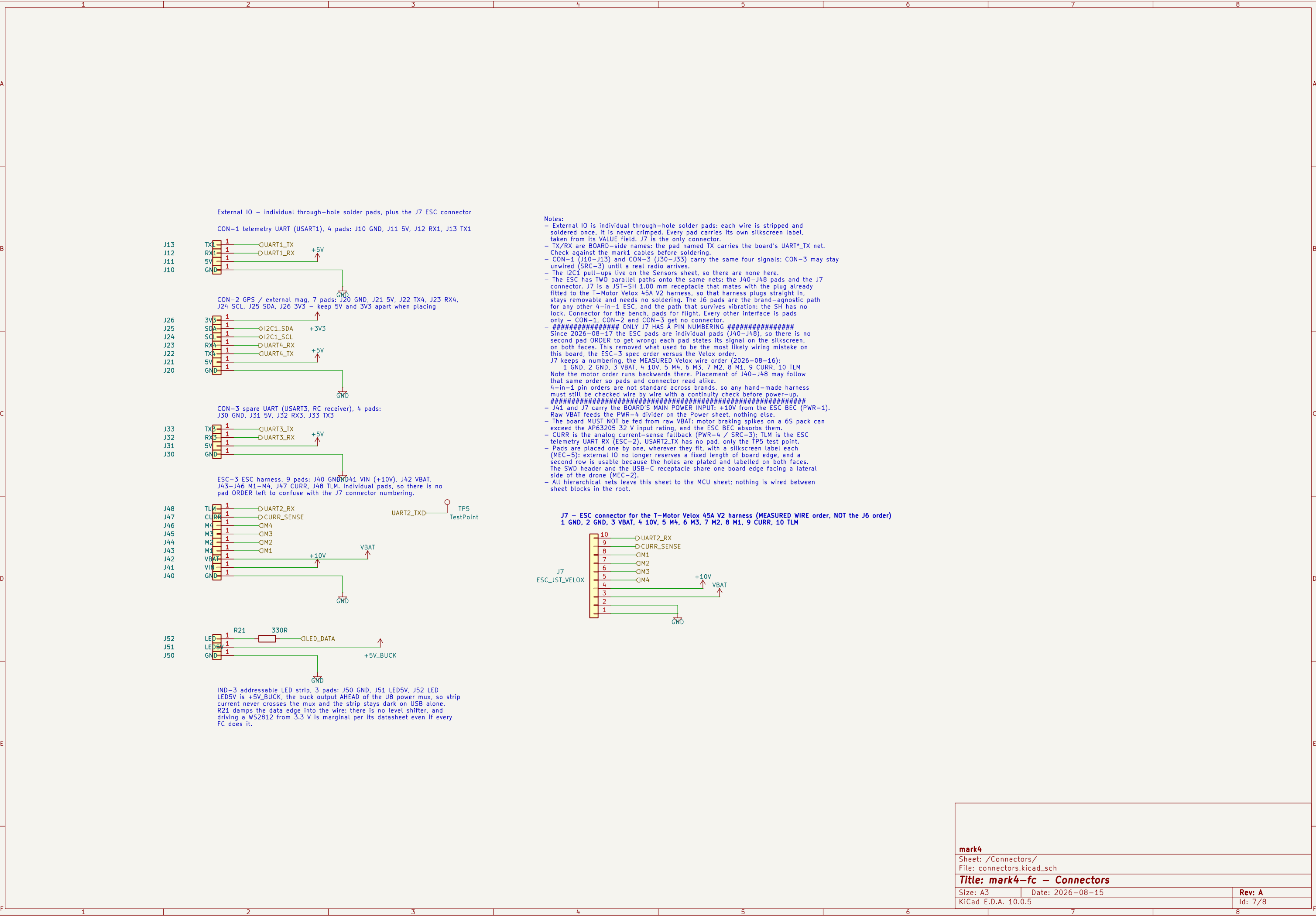
mark4

Sheet: /USB/
File: usb.kicad_sch

Title: mark4-fc – USB

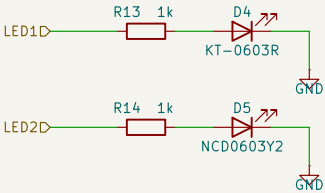
Size: A3 Date: 2026-08-15
KiCad E.D.A. 10.0.5

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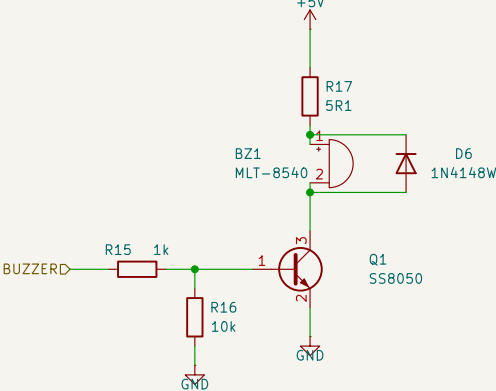


Indicators – status LEDs (IND-1) and buzzer driver (IND-2)

IND-1 – two status LEDs



IND-2 – buzzer, low-side driver



- Notes:
- IND-1: LED references chosen 2026-08-15. D4 = KT-0603R red (C2286, Basic), D5 = NCD0603Y2 yellow (C89811, Preferred-Extended, no feeder fee). R13 / R14 = 1k; at 3V3 that is about 1.0-1.7 mA depending on the LED Vf, which gives roughly 10-25 mcd – enough for an indoor status indicator.
 - IND-2: passive magnetic buzzer, PB5 = TIM3_CH2 so tones are free in firmware.
 - R17 = 5R1 (decided 2026-08-15): the MLT-8540 coil is a 3.6 V class part on a post-OR rail near 4.6 V, 5.1 ohm leaves the coil at about 3.37 V at the ~220 mA coil current, inside its 3.6 V rating, and also protects the coil from a firmware bug leaving PB5 high, which a duty cap would not. No fee-free 6.8 ohm 0805 exists at JLCPCB; 5R1 (C17724) is Basic and is the fee-free option with the lowest resistor dissipation. 0805 for the dissipation: about 0.25 W peak, so beeps must stay duty limited – a continuous tone would blow past the 125 mW rating of an 0805.
 - R15 = 1k base resistor gives deep saturation at ~220 mA.
 - R16 = 10k base-to-GND keeps the buzzer silent while PB5 is high-Z at reset.
 - D6 (1N4148W) is across the coil, cathode to the 5 V side.